

Purpose

Practice investment analyst work based on your Streamlit app: allocations, Sharpe ratio, Monte Carlo, efficient frontier, risk/return, scenario analysis, SQL, pivot tables, and

Important note

Data is synthetic training data modeled after the tickers and workflow in your app. For a real portfolio project, replace the Monthly_Returns sheet with yfinance/API export

Source / App logic

Based on user-provided Streamlit app: Investment Portfolio Analyzer Pro using yfinance tickers, Sharpe ratio, Monte Carlo simulation, random portfolio optimizer, efficient

How to use

1. Use Portfolio_Inputs to understand capital, risk-free rate, ticker allocation, and asset class mix.
2. Use Monthly_Returns and Risk_Metrics to analyze performance and risk.
3. Build pivot tables from Monthly_Returns, Risk_Metrics, Monte_Carlo_Scenarios, and Optimizer_Portfolios.
4. Answer SQL_Practice questions by importing the sheets into SQLite/Postgres/BigQuery.
5. Build charts and a dashboard using Dashboard_Tasks.
6. Write a short analyst recommendation: keep current allocation, increase/decrease risk, or use optimizer allocation.



Portfolio Inputs and Assumptions

Starting Capital (\$)	\$200,000
Risk-Free Rate	4.0%
Projection Years	10
Trading Days	252
Benchmark	SPY
Source Note	Synthetic training dataset based on user app defaults; replace with yfinance data for live portfolio.

Ticker	Asset Type
SPY	Equity ETF
QQQ	Growth ETF
AGG	Bond ETF
BIL	T-Bill ETF
VNQ	REIT ETF
VIG	Dividend ETF

Allocation %	Dollar Allocation	Investment Question
3000.0%	\$60,000	Should this allocation increase, decrease, or stay the same based on risk-adjusted return?
1500.0%	\$30,000	Should this allocation increase, decrease, or stay the same based on risk-adjusted return?
2000.0%	\$40,000	Should this allocation increase, decrease, or stay the same based on risk-adjusted return?
1500.0%	\$30,000	Should this allocation increase, decrease, or stay the same based on risk-adjusted return?
1000.0%	\$20,000	Should this allocation increase, decrease, or stay the same based on risk-adjusted return?
1000.0%	\$20,000	Should this allocation increase, decrease, or stay the same based on risk-adjusted return?



Month	SPY	QQQ	AGG	BIL	VNQ	VIG	Portfolio_Return	Market_Return_SPY
2024-01	6.9%	(0.5%)	2.1%	0.6%	(1.1%)	0.7%	2.4%	6.9%
2024-02	0.3%	(4.7%)	0.6%	0.2%	0.9%	(1.6%)	(0.6%)	0.3%
2024-03	1.7%	2.0%	1.5%	0.5%	0.3%	1.0%	1.3%	1.7%
2024-04	(7.6%)	(4.2%)	(0.2%)	0.3%	1.7%	(3.6%)	(3.1%)	(7.6%)
2024-05	0.2%	5.5%	1.0%	0.5%	(3.4%)	(0.6%)	0.8%	0.2%
2024-06	(1.7%)	2.3%	(0.7%)	0.4%	0.1%	1.3%	(0.1%)	(1.7%)
2024-07	(1.1%)	1.0%	(0.8%)	0.1%	(1.9%)	1.9%	(0.3%)	(1.1%)
2024-08	0.1%	2.5%	(1.6%)	0.3%	0.9%	(0.4%)	0.2%	0.1%
2024-09	1.3%	7.3%	1.3%	0.5%	7.5%	0.1%	2.6%	1.3%
2024-10	(2.9%)	(5.7%)	(2.4%)	0.3%	(0.2%)	(3.4%)	(2.5%)	(2.9%)
2024-11	(1.1%)	(2.2%)	0.8%	0.3%	(2.8%)	(3.8%)	(1.1%)	(1.1%)
2024-12	(0.2%)	1.2%	2.1%	0.6%	0.2%	1.7%	0.8%	(0.2%)
2025-01	3.3%	(3.3%)	(0.5%)	0.7%	4.9%	3.2%	1.3%	3.3%
2025-02	6.2%	8.2%	(0.1%)	0.4%	(1.7%)	2.5%	3.2%	6.2%
2025-03	(6.9%)	(6.2%)	(1.0%)	0.6%	(3.0%)	(4.6%)	(3.9%)	(6.9%)
2025-04	1.8%	(2.7%)	0.6%	0.0%	3.9%	(4.0%)	0.2%	1.8%
2025-05	1.9%	12.2%	(0.2%)	0.1%	5.3%	1.8%	3.1%	1.9%
2025-06	1.1%	2.5%	1.1%	0.6%	3.8%	2.5%	1.6%	1.1%
2025-07	0.2%	(1.2%)	(0.3%)	0.3%	(1.1%)	0.9%	(0.2%)	0.2%
2025-08	(0.6%)	5.4%	(0.4%)	0.4%	(1.3%)	(2.3%)	0.3%	(0.6%)
2025-09	6.6%	(0.1%)	2.4%	0.5%	(2.6%)	4.2%	2.7%	6.6%
2025-10	(3.1%)	(1.7%)	(1.0%)	0.3%	3.4%	4.5%	(0.5%)	(3.1%)
2025-11	3.2%	5.4%	0.8%	0.5%	3.4%	5.7%	2.9%	3.2%
2025-12	4.9%	7.7%	1.6%	0.5%	4.4%	7.4%	4.2%	4.9%

Excess_Return_vs_SPY	
(4.5%)	
(0.9%)	
(0.4%)	
4.5%	
0.5%	
1.6%	
0.8%	
0.0%	
1.3%	
0.4%	
(0.0%)	
1.0%	
(2.0%)	
(3.0%)	
3.0%	
(1.6%)	
1.2%	
0.6%	
(0.4%)	
0.9%	
(3.9%)	
2.5%	
(0.3%)	
(0.7%)	

Month	Total loss(Percentage)	Number of months
2024-02	(0.6%)	1
2024-04	(3.1%)	1
2024-06	(0.1%)	1
2024-07	(0.3%)	1
2024-10	(2.5%)	1
2024-11	(1.1%)	1
2025-03	(3.9%)	1
2025-07	(0.2%)	1
2025-10	(0.5%)	1
Grand Total	(12.3%)	9

Ticker	Asset Type	Allocation %	Annual Return	Annual Volatility	Sharpe Ratio	Max Drawdown	Portfolio Summary	
SPY	Equity ETF	3000.0%	6.7%	12.7%	0.21x	(12.5%)	Portfolio Annual Return	1.2%
QQQ	Growth ETF	1500.0%	15.0%	16.8%	0.65x	(11.0%)	Portfolio Annual Volatility	7.1%
AGG	Bond ETF	2000.0%	3.3%	4.3%	-0.16x	(4.1%)	Portfolio Sharpe Ratio	-0.39x
BIL	T-Bill ETF	1500.0%	5.0%	0.6%	1.56x	-	Risk-Free Rate	4.0%
VNQ	REIT ETF	1000.0%	10.8%	10.5%	0.65x	(5.1%)	Expected 10-Year Value	\$225,672
VIG	Dividend ETF	1000.0%	7.2%	11.1%	0.29x	(10.3%)	Worst Asset Drawdown	(12.5%)



Ticker	SPY	QQQ	AGG	BIL	VNQ	VIG	
SPY	1.00	0.48	0.57	0.20	0.14	0.60	
QQQ	0.48	1.00	0.26	0.00	0.30	0.50	
AGG	0.57	0.26	1.00	0.37	0.06	0.34	
BIL	0.20	0.00	0.37	1.00	-0.02	0.30	
VNQ	0.14	0.30	0.06	-0.02	1.00	0.34	
VIG	0.60	0.50	0.34	0.30	0.34	1.00	

Simulation_ID	Ending_Value_10Y	Monte Carlo KPI
1	\$456,592	5th Percentile Median Mean 95th Percentile
2	\$358,459	
3	\$387,049	
4	\$477,097	
5	\$500,560	Probability Ending Below Starting Capital
6	\$514,005	Probability Ending Above \$400k
7	\$325,644	Interpretation
8	\$624,136	
9	\$470,337	
10	\$644,290	
11	\$285,024	
12	\$418,916	
13	\$454,746	
14	\$394,633	
15	\$400,299	
16	\$385,379	
17	\$570,774	
18	\$434,767	
19	\$453,364	
20	\$485,699	
21	\$489,297	
22	\$372,539	
23	\$377,341	
24	\$405,208	
25	\$553,333	
26	\$384,007	
27	\$449,168	
28	\$403,396	
29	\$381,089	
30	\$336,874	
31	\$386,120	

32	\$472,188
33	\$397,781
34	\$423,715
35	\$462,535
36	\$611,140
37	\$336,415
38	\$681,711
39	\$636,133
40	\$383,479
41	\$221,071
42	\$390,418
43	\$408,450
44	\$539,549
45	\$335,548
46	\$497,734
47	\$392,057
48	\$479,684
49	\$328,633
50	\$336,704
51	\$504,246
52	\$342,629
53	\$375,803
54	\$322,880
55	\$520,972
56	\$582,438
57	\$404,473
58	\$320,180
59	\$450,142
60	\$339,130
61	\$346,490
62	\$540,024

63	\$329,337
64	\$462,737
65	\$463,985
66	\$450,929
67	\$375,591
68	\$411,266
69	\$432,447
70	\$444,575
71	\$448,796
72	\$359,162
73	\$579,689
74	\$410,567
75	\$286,802
76	\$500,178
77	\$497,965
78	\$449,671
79	\$329,574
80	\$405,845
81	\$503,955
82	\$395,022
83	\$202,092
84	\$353,969
85	\$554,590
86	\$386,600
87	\$428,154
88	\$521,518
89	\$383,651
90	\$441,055
91	\$420,157
92	\$453,709
93	\$261,237

94	\$400,820
95	\$356,689
96	\$626,999
97	\$385,260
98	\$535,536
99	\$312,193
100	\$548,921
101	\$425,602
102	\$325,329
103	\$406,733
104	\$475,158
105	\$344,797
106	\$621,757
107	\$432,959
108	\$464,543
109	\$386,597
110	\$330,477
111	\$261,763
112	\$312,597
113	\$423,737
114	\$454,404
115	\$291,572
116	\$446,684
117	\$457,802
118	\$246,582
119	\$564,452
120	\$409,732
121	\$412,076
122	\$581,574
123	\$469,887
124	\$453,751

125	\$418,883
126	\$557,559
127	\$456,033
128	\$243,315
129	\$557,637
130	\$553,113
131	\$475,100
132	\$486,247
133	\$352,152
134	\$318,183
135	\$547,017
136	\$461,057
137	\$376,456
138	\$441,892
139	\$368,823
140	\$432,777
141	\$363,118
142	\$480,713
143	\$385,532
144	\$475,128
145	\$392,538
146	\$578,736
147	\$389,592
148	\$317,202
149	\$608,912
150	\$452,859
151	\$415,026
152	\$395,297
153	\$404,816
154	\$539,295
155	\$437,228

156	\$346,021
157	\$576,102
158	\$340,461
159	\$355,348
160	\$517,068
161	\$391,495
162	\$282,832
163	\$424,537
164	\$311,238
165	\$346,150
166	\$416,016
167	\$400,445
168	\$403,358
169	\$352,267
170	\$350,779
171	\$522,973
172	\$439,751
173	\$420,242
174	\$414,099
175	\$315,478
176	\$368,841
177	\$438,534
178	\$431,040
179	\$532,141
180	\$495,267
181	\$537,749
182	\$482,270
183	\$452,064
184	\$243,498
185	\$407,370
186	\$445,495

187	\$437,447
188	\$443,518
189	\$406,833
190	\$534,689
191	\$554,575
192	\$443,514
193	\$316,456
194	\$487,058
195	\$361,259
196	\$542,756
197	\$511,660
198	\$648,350
199	\$488,520
200	\$504,531
201	\$564,565
202	\$334,276
203	\$374,732
204	\$250,546
205	\$411,980
206	\$490,237
207	\$302,685
208	\$436,701
209	\$368,888
210	\$347,827
211	\$407,822
212	\$497,378
213	\$559,296
214	\$333,823
215	\$420,348
216	\$465,320
217	\$515,168

218	\$459,847
219	\$332,342
220	\$444,516
221	\$507,929
222	\$342,378
223	\$410,504
224	\$462,371
225	\$445,947
226	\$408,967
227	\$274,086
228	\$341,123
229	\$479,110
230	\$324,033
231	\$434,114
232	\$236,437
233	\$294,321
234	\$429,563
235	\$398,530
236	\$435,501
237	\$610,712
238	\$543,317
239	\$591,881
240	\$373,157
241	\$502,215
242	\$404,191
243	\$393,719
244	\$558,629
245	\$382,589
246	\$413,588
247	\$338,218
248	\$280,014

249	\$462,567
250	\$299,742
251	\$332,503
252	\$381,204
253	\$344,462
254	\$667,158
255	\$421,802
256	\$330,530
257	\$333,514
258	\$501,964
259	\$359,477
260	\$450,061
261	\$397,855
262	\$428,649
263	\$321,647
264	\$371,847
265	\$400,577
266	\$497,641
267	\$376,614
268	\$281,724
269	\$489,419
270	\$433,182
271	\$491,696
272	\$451,280
273	\$507,109
274	\$456,661
275	\$540,885
276	\$290,036
277	\$358,423
278	\$277,322
279	\$419,599

280	\$423,108
281	\$568,190
282	\$457,261
283	\$526,422
284	\$415,291
285	\$269,388
286	\$565,336
287	\$403,490
288	\$590,485
289	\$374,857
290	\$424,981
291	\$417,355
292	\$348,202
293	\$382,987
294	\$485,997
295	\$494,810
296	\$375,591
297	\$355,426
298	\$327,825
299	\$366,480
300	\$449,390
301	\$232,204
302	\$501,224
303	\$376,669
304	\$496,231
305	\$324,635
306	\$528,415
307	\$299,113
308	\$395,891
309	\$395,940
310	\$529,311

311	\$489,204
312	\$319,178
313	\$300,267
314	\$303,980
315	\$551,760
316	\$279,520
317	\$427,261
318	\$523,566
319	\$538,697
320	\$492,748
321	\$503,427
322	\$377,472
323	\$454,886
324	\$611,069
325	\$356,408
326	\$290,719
327	\$346,096
328	\$528,842
329	\$356,432
330	\$441,989
331	\$409,007
332	\$400,501
333	\$362,310
334	\$309,553
335	\$433,934
336	\$602,597
337	\$297,592
338	\$464,239
339	\$387,994
340	\$322,767
341	\$623,076

342	\$515,694
343	\$557,580
344	\$593,490
345	\$203,732
346	\$459,807
347	\$469,572
348	\$327,274
349	\$326,933
350	\$601,290
351	\$385,001
352	\$470,831
353	\$420,464
354	\$532,376
355	\$312,046
356	\$431,253
357	\$420,404
358	\$434,750
359	\$393,616
360	\$434,175
361	\$418,428
362	\$429,341
363	\$697,838
364	\$496,485
365	\$325,970
366	\$491,203
367	\$588,537
368	\$347,810
369	\$656,315
370	\$375,225
371	\$423,957
372	\$360,324

373	\$378,569
374	\$389,042
375	\$452,054
376	\$467,258
377	\$409,934
378	\$423,761
379	\$518,575
380	\$554,529
381	\$460,790
382	\$405,948
383	\$361,601
384	\$382,720
385	\$348,663
386	\$394,935
387	\$470,556
388	\$532,571
389	\$366,704
390	\$436,476
391	\$377,300
392	\$439,380
393	\$502,022
394	\$333,365
395	\$376,242
396	\$455,531
397	\$417,605
398	\$524,854
399	\$400,518
400	\$826,796
401	\$309,868
402	\$437,444
403	\$400,302

404	\$290,033
405	\$574,334
406	\$422,663
407	\$298,385
408	\$428,630
409	\$346,757
410	\$334,513
411	\$472,560
412	\$449,091
413	\$352,711
414	\$297,979
415	\$573,984
416	\$389,702
417	\$412,423
418	\$318,616
419	\$322,737
420	\$418,828
421	\$682,893
422	\$406,856
423	\$399,406
424	\$285,285
425	\$455,912
426	\$449,202
427	\$507,200
428	\$486,527
429	\$495,429
430	\$357,195
431	\$360,369
432	\$498,158
433	\$411,878
434	\$386,041

435	\$439,151
436	\$368,055
437	\$324,500
438	\$259,433
439	\$356,852
440	\$403,933
441	\$479,044
442	\$258,204
443	\$397,365
444	\$482,328
445	\$342,512
446	\$313,002
447	\$371,923
448	\$514,325
449	\$475,543
450	\$506,035
451	\$312,801
452	\$438,808
453	\$477,083
454	\$306,769
455	\$439,521
456	\$543,053
457	\$457,992
458	\$494,484
459	\$501,111
460	\$402,508
461	\$354,363
462	\$447,287
463	\$305,005
464	\$324,902
465	\$354,868

466	\$339,719
467	\$343,222
468	\$348,508
469	\$419,659
470	\$269,607
471	\$375,072
472	\$453,767
473	\$266,340
474	\$438,879
475	\$310,624
476	\$368,847
477	\$479,338
478	\$372,263
479	\$419,771
480	\$420,973
481	\$461,452
482	\$425,207
483	\$357,399
484	\$398,860
485	\$437,129
486	\$535,925
487	\$360,090
488	\$365,625
489	\$364,878
490	\$405,067
491	\$660,809
492	\$339,615
493	\$634,922
494	\$416,612
495	\$424,109
496	\$527,627

497	\$502,250
498	\$388,833
499	\$353,578
500	\$365,388

Value

\$289,871

\$418,628

\$424,343

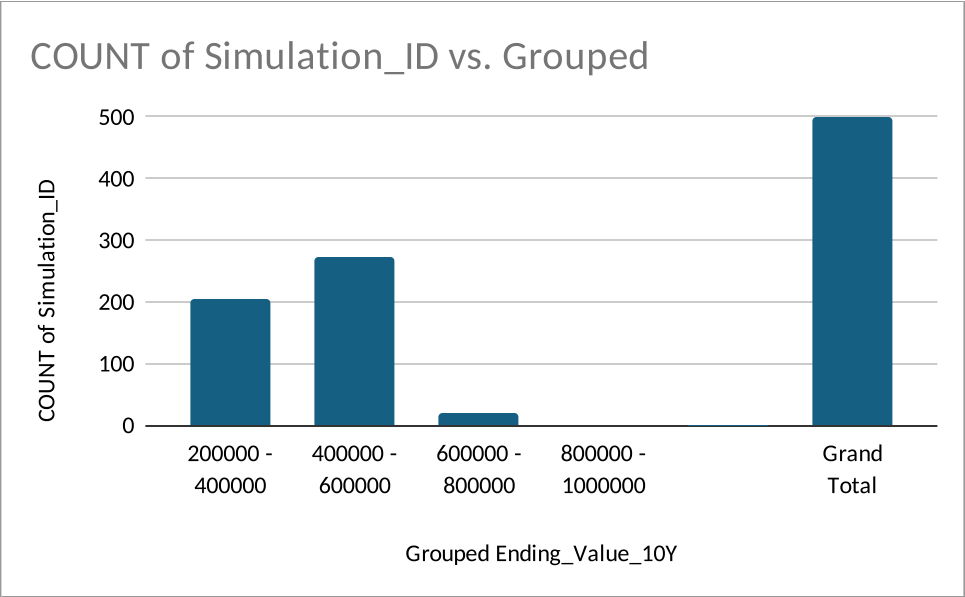
\$582,743

0.0%

59.0%

Use percentiles to explain downside risk and upside opportunity.

Grouped Ending_Value_10Y	COUNT of Simulation_ID	SUM of Simulation_ID
200000 - 400000	205	42.46%
400000 - 600000	274	53.50%
600000 - 800000	20	3.72%
800000 - 1000000	1	0.32%
	0	0.00%
Grand Total	500	100.00%



Portfolio_I					QQQ_Weigh	AGG_Weigh				
D	Return	Volatility	Sharpe	SPY_Weight	t	t	BIL_Weight	VNQ_Weight	VIG_Weight	
P001	7.5%	7.3%	0.48x	30.0%	17.6%	28.6%	11.7%	4.7%	7.4%	
P002	7.0%	5.2%	0.56x	23.4%	6.8%	22.8%	26.7%	16.1%	4.2%	
P003	7.3%	6.8%	0.48x	21.9%	13.3%	31.9%	7.8%	8.2%	16.9%	
P004	7.4%	5.3%	0.66x	21.9%	8.3%	16.6%	29.6%	20.9%	2.6%	
P005	8.3%	8.4%	0.52x	16.4%	16.5%	17.7%	0.9%	10.8%	37.6%	
P006	9.0%	7.4%	0.68x	20.8%	22.1%	17.8%	10.2%	22.0%	7.1%	
P007	8.6%	7.9%	0.58x	20.8%	20.6%	15.5%	8.8%	11.6%	22.6%	
P008	8.8%	6.9%	0.70x	10.6%	12.9%	13.2%	12.1%	30.9%	20.3%	
P009	8.8%	6.3%	0.77x	0.6%	26.1%	17.6%	27.4%	17.4%	10.9%	
P010	8.2%	6.6%	0.64x	30.5%	6.3%	10.1%	17.6%	31.3%	4.2%	
P011	8.8%	7.2%	0.66x	11.4%	17.9%	22.9%	3.5%	25.5%	18.8%	
P012	8.4%	5.8%	0.76x	15.8%	11.6%	9.6%	27.2%	30.4%	5.4%	
P013	7.9%	5.7%	0.68x	7.1%	22.4%	25.7%	27.6%	10.2%	7.0%	
P014	6.8%	4.4%	0.65x	3.6%	18.8%	28.0%	43.5%	1.6%	4.5%	
P015	9.4%	10.1%	0.53x	29.4%	30.2%	5.6%	4.3%	1.3%	29.2%	
P016	7.9%	6.0%	0.64x	14.5%	12.5%	34.4%	6.3%	27.7%	4.7%	
P017	8.3%	6.6%	0.64x	2.1%	21.6%	25.6%	15.0%	13.0%	22.7%	
P018	5.8%	3.5%	0.53x	3.4%	12.4%	40.2%	42.2%	1.7%	0.1%	
P019	7.6%	6.5%	0.55x	20.3%	11.9%	20.2%	17.0%	11.8%	18.8%	
P020	7.1%	5.5%	0.57x	17.8%	0.5%	13.7%	25.9%	22.9%	19.2%	
P021	9.3%	8.1%	0.65x	17.4%	22.8%	8.1%	11.0%	17.6%	23.0%	
P022	7.2%	6.6%	0.48x	16.7%	8.7%	14.1%	22.8%	5.1%	32.6%	
P023	7.3%	6.8%	0.49x	21.7%	8.3%	23.2%	10.2%	13.0%	23.7%	
P024	8.1%	6.4%	0.64x	21.0%	14.9%	22.3%	14.0%	21.3%	6.6%	
P025	8.2%	7.3%	0.57x	2.8%	22.1%	19.3%	17.0%	4.3%	34.5%	
P026	10.0%	8.4%	0.72x	3.1%	27.0%	8.9%	7.9%	26.3%	26.8%	
P027	9.4%	7.5%	0.72x	19.5%	26.5%	4.9%	23.8%	17.2%	8.2%	
P028	6.0%	5.2%	0.37x	24.2%	4.4%	32.1%	26.0%	4.3%	8.9%	
P029	9.5%	7.3%	0.75x	14.6%	23.8%	11.6%	14.9%	27.7%	7.4%	
P030	7.7%	7.4%	0.50x	23.3%	13.9%	12.6%	17.7%	4.9%	27.6%	
P031	8.2%	6.9%	0.61x	3.3%	15.9%	28.0%	4.0%	20.3%	28.3%	

P032	7.5%	6.3%	0.55x	22.6%	7.9%	14.2%	21.4%	14.2%	19.8%
P033	6.8%	5.4%	0.53x	24.2%	1.3%	12.0%	31.1%	16.3%	15.1%
P034	7.7%	5.5%	0.69x	12.4%	6.5%	27.3%	16.1%	35.9%	1.8%
P035	7.1%	6.1%	0.51x	25.4%	2.7%	23.9%	13.3%	21.5%	13.1%
P036	9.8%	8.0%	0.73x	5.7%	34.2%	23.2%	7.8%	21.5%	7.6%
P037	8.5%	8.0%	0.57x	17.0%	25.2%	25.8%	4.6%	8.1%	19.4%
P038	8.6%	6.8%	0.68x	15.6%	25.8%	17.4%	24.5%	11.1%	5.7%
P039	7.5%	5.2%	0.68x	11.7%	1.0%	15.3%	26.7%	35.4%	9.9%
P040	8.2%	6.3%	0.66x	17.1%	18.2%	16.7%	23.3%	14.0%	10.7%
P041	7.5%	6.1%	0.57x	34.6%	0.3%	0.4%	31.8%	24.2%	8.7%
P042	9.2%	6.9%	0.76x	12.9%	23.4%	19.1%	14.5%	28.1%	1.9%
P043	8.7%	7.1%	0.67x	20.7%	15.9%	9.8%	16.2%	22.0%	15.3%
P044	7.6%	6.0%	0.61x	19.2%	20.2%	26.3%	25.6%	8.0%	0.6%
P045	9.2%	6.6%	0.78x	11.2%	14.7%	10.0%	18.5%	37.9%	7.7%
P046	9.8%	8.5%	0.68x	12.6%	27.6%	4.3%	11.7%	18.3%	25.5%
P047	6.5%	6.3%	0.39x	30.8%	9.2%	34.1%	15.6%	4.7%	5.7%
P048	10.1%	8.8%	0.70x	13.8%	25.7%	2.2%	6.9%	25.4%	26.0%
P049	5.7%	4.4%	0.38x	8.8%	2.5%	28.7%	34.9%	2.2%	23.0%
P050	9.9%	8.9%	0.66x	5.7%	24.3%	4.5%	5.3%	21.8%	38.3%
P051	6.8%	3.8%	0.74x	1.7%	6.0%	12.2%	49.2%	14.5%	16.4%
P052	9.5%	8.0%	0.69x	14.3%	28.4%	9.3%	15.7%	15.8%	16.6%
P053	9.1%	8.5%	0.60x	15.1%	27.2%	6.8%	17.1%	4.4%	29.5%
P054	9.8%	7.9%	0.73x	12.1%	28.4%	18.9%	5.8%	26.4%	8.4%
P055	10.1%	9.2%	0.66x	25.8%	30.9%	4.2%	7.4%	17.2%	14.5%
P056	7.5%	5.1%	0.68x	11.1%	9.5%	12.7%	34.5%	16.1%	16.1%
P057	8.0%	5.1%	0.79x	9.5%	7.2%	9.8%	33.5%	33.6%	6.5%
P058	8.2%	8.1%	0.52x	15.3%	22.4%	21.4%	7.8%	2.4%	30.8%
P059	7.4%	5.7%	0.60x	4.1%	12.5%	24.0%	21.5%	12.8%	25.1%
P060	7.5%	7.0%	0.49x	23.3%	11.3%	20.8%	12.2%	9.1%	23.2%
P061	7.3%	6.8%	0.48x	25.1%	4.4%	17.6%	12.3%	16.1%	24.5%
P062	9.3%	7.4%	0.71x	10.4%	18.5%	7.3%	13.7%	27.4%	22.7%

P063	8.2%	5.9%	0.70x	12.1%	9.3%	20.1%	15.8%	32.9%	9.9%
P064	6.6%	5.9%	0.44x	26.7%	6.8%	23.2%	25.0%	5.5%	12.8%
P065	8.4%	6.6%	0.68x	26.6%	15.4%	4.0%	28.8%	18.2%	6.9%
P066	8.3%	5.6%	0.78x	2.8%	28.1%	15.8%	43.9%	8.5%	0.9%
P067	7.8%	6.6%	0.57x	19.9%	9.1%	18.7%	12.5%	19.3%	20.5%
P068	8.0%	5.9%	0.68x	2.7%	8.9%	5.9%	30.6%	22.2%	29.6%
P069	8.0%	6.7%	0.59x	19.6%	15.8%	17.4%	18.7%	12.1%	16.3%
P070	7.9%	7.7%	0.50x	36.2%	17.5%	8.0%	24.7%	2.1%	11.5%
P071	10.1%	9.2%	0.66x	15.1%	32.2%	1.0%	12.9%	12.3%	26.5%
P072	7.7%	6.7%	0.56x	25.4%	13.9%	24.5%	12.3%	14.6%	9.3%
P073	9.5%	7.8%	0.70x	6.8%	30.7%	20.8%	9.2%	18.6%	14.0%
P074	7.9%	6.2%	0.62x	1.3%	25.1%	31.2%	21.9%	5.3%	15.1%
P075	7.8%	5.6%	0.67x	6.4%	5.3%	9.2%	28.0%	26.3%	24.8%
P076	7.1%	9.6%	0.33x	47.7%	5.5%	11.3%	2.2%	0.2%	33.0%
P077	7.8%	7.1%	0.53x	12.8%	17.7%	27.2%	9.0%	7.7%	25.7%
P078	11.5%	10.7%	0.70x	1.5%	56.8%	21.3%	2.1%	13.3%	5.0%
P079	7.1%	6.9%	0.44x	40.0%	2.6%	2.5%	29.1%	9.9%	15.8%
P080	7.4%	7.2%	0.47x	32.5%	17.6%	30.1%	11.6%	5.4%	2.8%
P081	8.2%	8.3%	0.51x	34.8%	5.1%	3.3%	6.5%	21.7%	28.6%
P082	8.5%	6.3%	0.72x	15.8%	19.8%	15.9%	23.7%	19.2%	5.6%
P083	9.0%	7.8%	0.64x	10.2%	30.1%	13.6%	20.5%	5.2%	20.5%
P084	6.7%	6.2%	0.43x	21.7%	5.0%	29.3%	12.3%	10.8%	21.1%
P085	9.1%	8.7%	0.59x	26.4%	24.1%	16.1%	1.6%	14.0%	17.8%
P086	7.6%	6.0%	0.61x	29.0%	7.0%	16.2%	22.3%	23.9%	1.6%
P087	6.8%	5.5%	0.50x	22.7%	2.9%	22.1%	22.4%	15.9%	14.0%
P088	7.3%	5.3%	0.61x	11.1%	12.4%	29.3%	20.9%	14.8%	11.4%
P089	8.0%	7.0%	0.57x	20.0%	18.8%	18.6%	18.6%	7.2%	16.8%
P090	8.6%	6.7%	0.68x	15.1%	20.1%	16.2%	18.3%	17.6%	12.7%
P091	6.3%	5.7%	0.40x	29.2%	3.2%	16.1%	33.7%	3.9%	13.8%
P092	9.5%	8.4%	0.66x	24.7%	17.8%	2.9%	6.8%	26.3%	21.6%
P093	8.9%	7.3%	0.67x	14.2%	22.0%	22.6%	7.1%	21.4%	12.7%

P094	7.5%	5.4%	0.65x	18.6%	2.9%	11.6%	27.5%	27.2%	12.0%
P095	8.5%	7.1%	0.63x	7.5%	29.2%	15.3%	28.8%	1.1%	18.1%
P096	6.3%	3.9%	0.61x	11.8%	0.2%	20.4%	39.6%	19.2%	8.9%
P097	8.9%	7.7%	0.64x	16.9%	21.1%	18.3%	6.0%	18.8%	18.8%
P098	9.8%	9.3%	0.63x	16.6%	27.2%	1.3%	7.4%	14.0%	33.4%
P099	7.5%	7.0%	0.51x	18.0%	17.3%	21.3%	18.7%	1.6%	23.1%
P100	6.8%	7.2%	0.39x	26.7%	7.5%	20.8%	15.8%	1.0%	28.1%
P101	7.8%	5.7%	0.68x	9.3%	17.0%	22.2%	25.2%	14.2%	12.1%
P102	7.1%	5.0%	0.62x	0.6%	6.6%	42.1%	10.6%	30.5%	9.6%
P103	7.4%	6.1%	0.56x	23.8%	6.3%	23.2%	13.4%	22.5%	10.8%
P104	8.1%	7.8%	0.53x	29.6%	15.4%	12.4%	12.7%	9.3%	20.6%
P105	7.3%	5.3%	0.63x	7.3%	6.4%	40.2%	8.8%	33.8%	3.5%
P106	7.8%	6.2%	0.61x	29.6%	5.2%	17.9%	16.6%	30.5%	0.2%
P107	8.3%	6.4%	0.67x	7.1%	19.4%	19.4%	20.2%	14.7%	19.3%
P108	7.0%	6.7%	0.45x	14.3%	10.4%	26.6%	13.0%	4.3%	31.4%
P109	9.0%	7.1%	0.71x	3.6%	21.7%	5.9%	24.3%	17.7%	26.7%
P110	8.4%	7.1%	0.62x	14.5%	20.2%	11.9%	21.4%	9.8%	22.3%
P111	10.5%	8.6%	0.75x	20.6%	29.3%	1.4%	11.6%	28.7%	8.5%
P112	10.0%	9.2%	0.65x	28.9%	22.3%	0.5%	2.2%	26.0%	20.1%
P113	8.4%	8.8%	0.50x	30.7%	11.3%	5.5%	4.1%	14.2%	34.1%
P114	8.4%	6.7%	0.66x	26.6%	15.9%	15.4%	18.4%	22.0%	1.7%
P115	9.1%	8.2%	0.62x	25.7%	24.0%	12.8%	9.2%	14.5%	13.8%
P116	6.9%	4.5%	0.64x	10.6%	4.2%	24.2%	29.7%	21.7%	9.7%
P117	6.8%	6.1%	0.46x	15.5%	6.7%	33.3%	8.9%	12.5%	23.0%
P118	9.2%	7.5%	0.69x	26.0%	15.9%	4.9%	14.5%	28.8%	9.9%
P119	8.0%	7.1%	0.57x	15.2%	10.1%	13.7%	12.5%	17.8%	30.8%
P120	8.1%	7.4%	0.55x	27.3%	5.8%	7.9%	11.5%	22.8%	24.6%
P121	7.6%	6.1%	0.59x	20.9%	6.2%	6.4%	28.1%	16.7%	21.7%
P122	8.9%	7.5%	0.66x	7.0%	21.3%	10.1%	16.7%	15.3%	29.6%
P123	9.5%	7.1%	0.78x	11.6%	23.8%	9.9%	19.5%	27.6%	7.5%
P124	8.6%	6.5%	0.70x	10.2%	23.9%	20.3%	21.7%	14.7%	9.2%

P125	9.4%	7.2%	0.75x	3.6%	22.0%	2.9%	23.7%	23.6%	24.1%
P126	8.4%	6.8%	0.65x	20.8%	17.6%	10.3%	22.2%	15.5%	13.6%
P127	7.6%	6.0%	0.61x	6.1%	5.4%	30.6%	6.3%	31.9%	19.7%
P128	7.4%	5.1%	0.66x	4.1%	18.0%	25.4%	31.8%	8.0%	12.6%
P129	8.5%	8.2%	0.55x	22.0%	22.1%	10.7%	15.0%	4.1%	26.0%
P130	6.7%	6.4%	0.42x	32.9%	4.1%	23.8%	17.4%	10.8%	11.0%
P131	7.5%	6.2%	0.57x	29.3%	3.2%	17.5%	15.9%	28.0%	6.0%
P132	10.0%	8.7%	0.69x	1.5%	44.4%	16.3%	20.5%	6.0%	11.3%
P133	10.3%	8.9%	0.71x	11.1%	33.5%	6.6%	9.4%	19.7%	19.7%
P134	8.9%	6.5%	0.75x	1.3%	26.5%	22.4%	20.7%	18.6%	10.5%
P135	8.4%	6.6%	0.67x	14.5%	18.2%	15.6%	19.1%	16.8%	15.8%
P136	8.7%	8.4%	0.56x	29.8%	26.0%	10.9%	16.5%	3.1%	13.7%
P137	7.1%	6.6%	0.47x	27.1%	9.2%	24.5%	14.1%	9.8%	15.3%
P138	8.1%	6.3%	0.65x	16.9%	13.2%	12.7%	22.1%	18.3%	16.7%
P139	7.8%	7.0%	0.53x	5.4%	0.9%	17.4%	5.7%	30.6%	40.0%
P140	7.5%	6.2%	0.57x	19.1%	11.5%	21.7%	18.5%	13.5%	15.7%
P141	6.2%	4.2%	0.52x	5.7%	0.6%	28.9%	30.4%	16.0%	18.2%
P142	6.4%	5.7%	0.42x	24.6%	3.1%	22.5%	24.8%	7.4%	17.6%
P143	9.6%	8.6%	0.65x	22.0%	25.9%	11.3%	4.2%	19.9%	16.8%
P144	8.4%	5.8%	0.76x	21.8%	17.3%	1.2%	40.7%	16.9%	2.0%
P145	9.9%	6.7%	0.88x	4.6%	26.3%	1.6%	34.1%	33.1%	0.2%
P146	7.2%	6.2%	0.51x	24.9%	11.8%	11.2%	33.1%	2.8%	16.1%
P147	6.7%	7.2%	0.37x	38.6%	3.5%	24.6%	10.4%	9.3%	13.6%
P148	7.8%	5.8%	0.65x	16.9%	9.9%	17.0%	21.8%	21.7%	12.7%
P149	9.3%	8.6%	0.62x	9.9%	24.5%	3.6%	13.3%	11.7%	37.0%
P150	8.3%	7.1%	0.60x	17.8%	12.4%	9.0%	16.8%	16.8%	27.2%
P151	10.1%	8.5%	0.72x	17.4%	34.1%	1.8%	21.2%	14.8%	10.6%
P152	9.1%	7.0%	0.72x	13.0%	26.6%	10.6%	25.1%	14.2%	10.5%
P153	6.3%	5.1%	0.45x	4.0%	4.9%	47.6%	7.3%	15.9%	20.3%
P154	7.6%	6.4%	0.56x	4.2%	12.2%	13.4%	25.5%	8.2%	36.4%
P155	9.8%	8.5%	0.68x	19.0%	28.1%	8.2%	9.8%	18.6%	16.2%

P156	7.2%	6.4%	0.51x	12.9%	5.6%	19.9%	14.8%	15.1%	31.7%
P157	5.8%	4.5%	0.41x	2.9%	10.4%	52.0%	18.8%	3.4%	12.5%
P158	10.1%	9.8%	0.62x	3.0%	39.9%	12.5%	6.8%	4.1%	33.6%
P159	9.3%	8.0%	0.66x	16.3%	26.3%	11.4%	12.3%	15.3%	18.4%
P160	7.3%	5.2%	0.63x	15.8%	7.7%	17.1%	28.9%	17.1%	13.4%
P161	11.2%	10.9%	0.66x	3.1%	42.0%	0.9%	0.3%	14.3%	39.4%
P162	8.9%	7.3%	0.67x	10.1%	26.7%	25.4%	10.3%	16.4%	11.0%
P163	8.3%	7.0%	0.60x	20.8%	16.5%	16.4%	14.6%	14.9%	16.8%
P164	8.5%	7.7%	0.58x	26.4%	8.9%	5.0%	10.1%	23.7%	25.9%
P165	8.0%	5.9%	0.68x	10.7%	14.5%	20.9%	20.0%	20.3%	13.7%
P166	7.6%	7.2%	0.50x	34.8%	7.0%	27.7%	0.8%	24.1%	5.6%
P167	8.0%	7.2%	0.56x	21.7%	20.6%	14.6%	22.8%	3.2%	17.1%
P168	8.3%	6.0%	0.72x	9.6%	16.4%	17.8%	21.6%	22.0%	12.5%
P169	8.0%	6.4%	0.63x	25.4%	11.8%	20.0%	14.7%	23.9%	4.3%
P170	8.9%	8.0%	0.61x	19.2%	21.4%	18.2%	4.1%	16.2%	20.9%
P171	11.1%	9.2%	0.77x	7.6%	42.6%	13.6%	7.4%	27.8%	1.1%
P172	7.6%	6.2%	0.58x	21.6%	15.4%	34.2%	10.5%	17.5%	0.8%
P173	9.7%	7.8%	0.73x	15.8%	23.2%	6.4%	12.4%	26.9%	15.3%
P174	7.9%	5.2%	0.75x	12.8%	7.6%	0.9%	39.9%	23.8%	15.0%
P175	8.3%	7.1%	0.61x	2.6%	21.1%	25.6%	9.7%	12.7%	28.2%
P176	7.9%	6.6%	0.59x	9.4%	12.9%	13.7%	20.0%	12.9%	31.0%
P177	8.2%	6.2%	0.67x	24.8%	8.9%	2.6%	28.5%	23.9%	11.4%
P178	7.5%	6.2%	0.57x	21.6%	5.8%	11.3%	23.1%	17.9%	20.2%
P179	8.4%	6.1%	0.73x	16.4%	12.7%	14.2%	20.6%	29.5%	6.7%
P180	8.2%	7.2%	0.59x	25.2%	8.4%	14.0%	8.0%	25.9%	18.5%
P181	8.9%	7.1%	0.68x	10.2%	26.1%	18.7%	17.3%	14.0%	13.8%
P182	7.7%	6.5%	0.56x	33.0%	9.0%	6.1%	28.1%	13.9%	9.8%
P183	7.6%	5.1%	0.70x	8.3%	3.4%	16.4%	26.7%	32.6%	12.7%
P184	7.7%	7.3%	0.50x	3.6%	13.9%	7.9%	25.4%	2.0%	47.2%
P185	7.8%	6.1%	0.63x	15.9%	21.8%	25.7%	25.0%	8.3%	3.4%
P186	8.3%	6.8%	0.63x	10.1%	23.5%	23.9%	16.8%	9.7%	15.9%

P187	9.5%	8.3%	0.67x	12.9%	5.9%	1.2%	1.7%	48.2%	30.1%
P188	9.5%	8.0%	0.69x	12.8%	26.6%	19.0%	4.5%	22.8%	14.3%
P189	9.5%	7.9%	0.69x	21.7%	33.7%	1.8%	32.8%	6.4%	3.6%
P190	10.1%	8.7%	0.70x	11.4%	27.9%	10.9%	1.8%	26.1%	21.9%
P191	8.5%	7.0%	0.64x	18.4%	18.6%	23.2%	8.1%	20.7%	11.1%
P192	6.0%	4.5%	0.45x	19.4%	1.6%	35.3%	25.4%	15.5%	2.8%
P193	7.3%	5.0%	0.66x	9.4%	13.0%	23.4%	29.6%	13.6%	11.0%
P194	9.7%	7.1%	0.81x	1.7%	29.0%	7.2%	27.1%	23.4%	11.5%
P195	8.1%	5.9%	0.69x	7.6%	4.3%	6.8%	26.3%	32.9%	22.2%
P196	8.5%	6.5%	0.69x	6.7%	21.1%	16.5%	22.4%	14.3%	19.1%
P197	7.9%	4.7%	0.84x	9.5%	9.6%	4.3%	44.2%	25.9%	6.6%
P198	8.7%	9.8%	0.48x	39.8%	21.5%	7.6%	2.8%	3.6%	24.7%
P199	8.4%	7.1%	0.62x	10.1%	23.0%	21.2%	15.2%	9.9%	20.6%
P200	7.5%	6.3%	0.56x	21.9%	12.9%	18.2%	21.9%	10.3%	14.8%
P201	9.5%	7.8%	0.70x	8.5%	17.8%	11.0%	5.1%	32.5%	25.1%
P202	8.5%	7.6%	0.59x	17.8%	25.2%	22.7%	11.5%	8.0%	14.8%
P203	8.6%	7.4%	0.63x	22.5%	24.6%	24.7%	10.5%	14.6%	3.1%
P204	10.5%	9.6%	0.68x	21.8%	38.1%	8.1%	6.5%	16.5%	9.0%
P205	8.6%	6.6%	0.70x	9.8%	5.2%	22.1%	8.9%	52.7%	1.2%
P206	6.4%	3.6%	0.66x	1.2%	11.1%	37.3%	36.7%	11.4%	2.3%
P207	7.1%	6.0%	0.52x	24.9%	8.9%	30.6%	13.5%	16.0%	6.1%
P208	8.3%	6.9%	0.63x	21.4%	24.2%	21.9%	20.3%	10.3%	2.0%
P209	8.3%	6.7%	0.65x	1.0%	23.0%	34.1%	7.2%	16.5%	18.1%
P210	8.3%	7.9%	0.55x	34.3%	3.8%	2.7%	9.1%	27.9%	22.3%
P211	7.4%	8.0%	0.42x	28.1%	15.2%	25.9%	5.4%	0.6%	24.7%
P212	6.7%	6.1%	0.44x	15.9%	3.5%	20.1%	20.0%	9.5%	31.1%
P213	9.1%	7.6%	0.66x	26.5%	14.6%	7.6%	9.5%	28.9%	13.0%
P214	7.4%	5.2%	0.65x	16.2%	4.4%	16.5%	27.1%	24.8%	10.9%
P215	7.7%	6.5%	0.58x	8.7%	19.6%	19.1%	25.7%	2.8%	24.1%
P216	7.9%	6.6%	0.59x	32.1%	5.6%	13.6%	15.8%	28.3%	4.6%
P217	7.1%	6.2%	0.50x	0.8%	14.2%	37.3%	10.3%	7.0%	30.5%

P218	9.4%	7.0%	0.77x	1.3%	24.5%	16.5%	15.4%	28.8%	13.5%
P219	7.3%	4.9%	0.67x	15.3%	8.4%	25.8%	25.2%	23.8%	1.5%
P220	7.5%	6.0%	0.58x	13.7%	15.4%	15.9%	30.1%	5.2%	19.7%
P221	9.2%	8.1%	0.64x	29.6%	15.4%	10.3%	2.6%	29.2%	12.9%
P222	7.8%	5.9%	0.63x	12.1%	8.5%	9.4%	26.8%	18.6%	24.6%
P223	9.2%	7.3%	0.72x	13.1%	24.0%	25.8%	4.9%	28.8%	3.3%
P224	8.0%	6.8%	0.58x	17.3%	11.9%	21.5%	9.2%	19.1%	20.9%
P225	8.7%	7.0%	0.67x	3.8%	32.8%	29.2%	20.0%	8.1%	6.2%
P226	8.1%	6.5%	0.63x	22.0%	13.8%	26.6%	8.5%	24.2%	4.8%
P227	6.3%	4.2%	0.55x	9.7%	2.8%	29.4%	30.7%	15.4%	12.0%
P228	8.5%	7.5%	0.60x	24.9%	13.6%	5.3%	16.9%	16.6%	22.7%
P229	8.9%	7.7%	0.64x	18.8%	25.3%	5.7%	24.1%	7.8%	18.3%
P230	9.9%	8.0%	0.74x	24.5%	23.6%	2.1%	14.2%	28.8%	6.9%
P231	8.4%	6.7%	0.66x	15.8%	12.7%	21.0%	8.0%	28.2%	14.4%
P232	9.2%	6.6%	0.79x	1.6%	27.7%	21.7%	21.2%	23.7%	4.2%
P233	6.6%	5.9%	0.44x	6.2%	2.0%	11.2%	32.4%	6.4%	41.8%
P234	8.7%	7.4%	0.64x	13.7%	24.0%	16.6%	15.6%	11.1%	19.0%
P235	10.6%	9.1%	0.72x	17.8%	35.3%	11.6%	3.9%	24.4%	7.0%
P236	9.2%	7.8%	0.67x	5.7%	31.2%	24.0%	10.2%	13.0%	15.9%
P237	8.1%	8.0%	0.51x	13.7%	14.2%	11.1%	10.6%	8.4%	42.0%
P238	9.1%	8.5%	0.60x	19.3%	24.9%	15.8%	4.4%	12.6%	23.0%
P239	10.1%	8.5%	0.71x	18.7%	25.0%	7.5%	4.3%	29.5%	15.0%
P240	8.4%	7.0%	0.64x	28.7%	18.3%	10.3%	22.8%	15.1%	4.9%
P241	8.6%	7.1%	0.65x	16.0%	20.4%	17.4%	13.8%	16.1%	16.3%
P242	12.6%	10.8%	0.80x	0.1%	50.8%	1.7%	3.8%	35.7%	7.9%
P243	8.9%	8.6%	0.57x	20.7%	26.9%	19.8%	3.7%	7.6%	21.5%
P244	9.0%	8.3%	0.60x	27.1%	21.9%	12.3%	7.4%	14.7%	16.5%
P245	10.1%	8.5%	0.71x	12.0%	31.2%	7.9%	10.6%	20.5%	17.7%
P246	7.4%	5.6%	0.60x	13.0%	1.2%	13.4%	23.4%	26.2%	22.7%
P247	9.3%	7.1%	0.74x	10.6%	19.5%	11.2%	14.2%	28.9%	15.6%
P248	7.3%	6.7%	0.49x	25.7%	17.8%	24.3%	21.7%	1.1%	9.3%

P249	7.6%	7.9%	0.46x	24.5%	17.9%	27.3%	5.6%	1.8%	23.0%
P250	8.5%	7.2%	0.63x	14.6%	15.2%	19.5%	5.9%	23.0%	21.8%
Current	7.7%	7.1%	0.53x	30.0%	15.0%	20.0%	15.0%	10.0%	10.0%

Optimizer K Formula

Best Sharpe	0.88x
Min Volatili	3.5%
Highest Ret	12.6%
Current Sharpe	0.53x

Exercise #	Data Sheet	Rows	Columns	Values
1	Portfolio_Inputs	Asset Type	SUM	Dollar Allocation
2	Risk_Metrics	Asset Type	AVERAGE	Sharpe Ratio
3	Risk_Metrics	Ticker	MAX	Annual Volatility
4	Monthly_Returns	Month	Portfolio_Return and Market_Return_SPY	
5	Monthly_Returns	Month	AVERAGE	Excess_Return_vs_SPY
6	Monte_Carlo_Scenarios	Ending Value Buckets	COUNT	Simulation_ID
7	Optimizer_Portfolios	Portfolio_ID	Return, Volatility, Sharpe	
8	Optimizer_Portfolios	Portfolio_ID	MAX	Sharpe
9	Risk_Metrics	Ticker	Annual Return and Annual Volatility	
10	Monthly_Returns	Month	COUNT	months with negative portfolio return

Business Question	
How is capital split across asset classes?	
Which asset type has best risk-adjusted performance?	
Which holdings are riskiest?	
How did portfolio returns compare to SPY over time?	
Which months outperformed the benchmark?	
What is the distribution of possible 10-year outcomes?	
Which optimizer portfolios dominate current portfolio?	
Find highest Sharpe allocation.	
Create risk-return comparison table.	
How often did the portfolio lose money?	

#	Question
1	Average annual return by asset type
2	Highest Sharpe holdings
3	Riskier than portfolio volatility
4	Monthly portfolio outperformance vs SPY
5	Count positive and negative months
6	Worst 5 monthly portfolio returns
7	Monte Carlo downside probability
8	Monte Carlo percentile buckets
9	Optimizer portfolios beating current Sharpe
10	Efficient frontier candidates
11	Allocation concentration in optimizer
12	Find portfolios with at least 30% bonds/cash

SQL Skeleton / Solution

```
SELECT asset_type, AVG(annual_return) AS avg_return FROM risk_metrics GROUP BY asset_type ORDER BY avg_return DESC;
```

```
SELECT ticker, sharpe_ratio FROM risk_metrics ORDER BY sharpe_ratio DESC;
```

```
SELECT ticker, annual_volatility FROM risk_metrics WHERE annual_volatility > (SELECT portfolio_annual_volatility FROM portfolio_summary);
```

```
SELECT month, portfolio_return, market_return_spy, excess_return_vs_spy FROM monthly_returns WHERE excess_return_vs_spy > 0 ORDER BY month;
```

```
SELECT CASE WHEN portfolio_return >= 0 THEN 'Positive' ELSE 'Negative' END AS month_type, COUNT(*) AS months FROM monthly_returns GROUP BY month_type;
```

```
SELECT month, portfolio_return FROM monthly_returns ORDER BY portfolio_return ASC LIMIT 5;
```

```
SELECT AVG(CASE WHEN ending_value_10y < 200000 THEN 1.0 ELSE 0 END) AS prob_loss FROM monte_carlo_scenarios;
```

```
SELECT CASE WHEN ending_value_10y < 250000 THEN '<250k' WHEN ending_value_10y < 400000 THEN '250k-400k' ELSE '400k+' END AS bucket, COUNT(*) FROM monte_carlo_scenarios;
```

```
SELECT * FROM optimizer_portfolios WHERE sharpe > (SELECT sharpe FROM optimizer_portfolios WHERE portfolio_id='Current') ORDER BY sharpe DESC;
```

```
SELECT portfolio_id, return, volatility, sharpe FROM optimizer_portfolios WHERE return > 0.08 AND volatility < 0.12 ORDER BY sharpe DESC;
```

```
SELECT portfolio_id, spy_weight, qqq_weight, agg_weight, bil_weight, vnq_weight, vig_weight FROM optimizer_portfolios ORDER BY sharpe DESC LIMIT 10;
```

```
SELECT * FROM optimizer_portfolios WHERE agg_weight + bil_weight >= 0.30 ORDER BY sharpe DESC;
```



Task	Chart/Table	Data Source	What to Explain
Portfolio allocation by ticker	Bar or pie chart	Portfolio_Inputs	Where the \$200k is invested.
Portfolio return vs SPY	Line chart	Monthly_Returns	Whether the portfolio beats the market and when.
Risk-return by ticker	Scatterplot	Risk_Metrics	Which holdings offer return for risk.
Sharpe ratio ranking	Bar chart	Risk_Metrics	Best risk-adjusted assets.
Correlation heatmap/table	Formatted table	Correlation_Data	Diversification quality.
Monte Carlo outcome distribution	Histogram/bar chart	Monte_Carlo_Scenarios	Downside/upside range of future wealth.
Efficient frontier	Scatterplot	Optimizer_Portfolios	Return-volatility tradeoff.
Recommended allocation	Table	Optimizer_Portfolios	Compare current vs highest Sharpe allocation.
Executive summary	Text box	All sheets	Recommendation: aggressive, balanced, defensive, or optimized.



Starting Capital

Portfolio Annual Return

Portfolio Annual Volatility

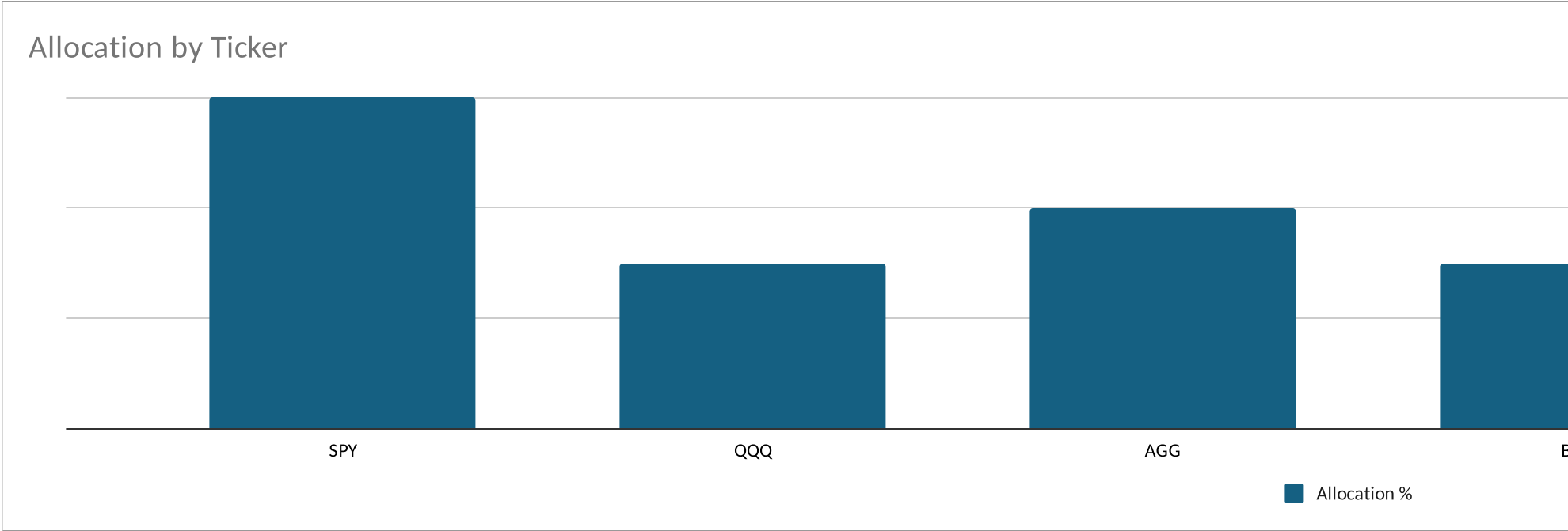
Sharpe Ratio

Expected 10-Year Value

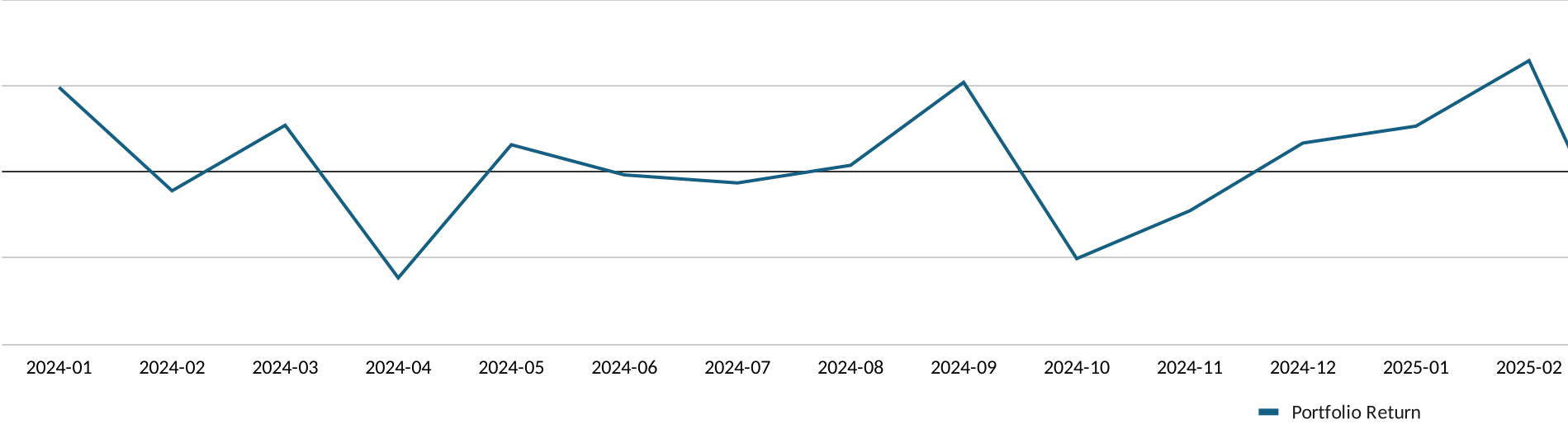
Downside Probability

Analyst Recommendation Template

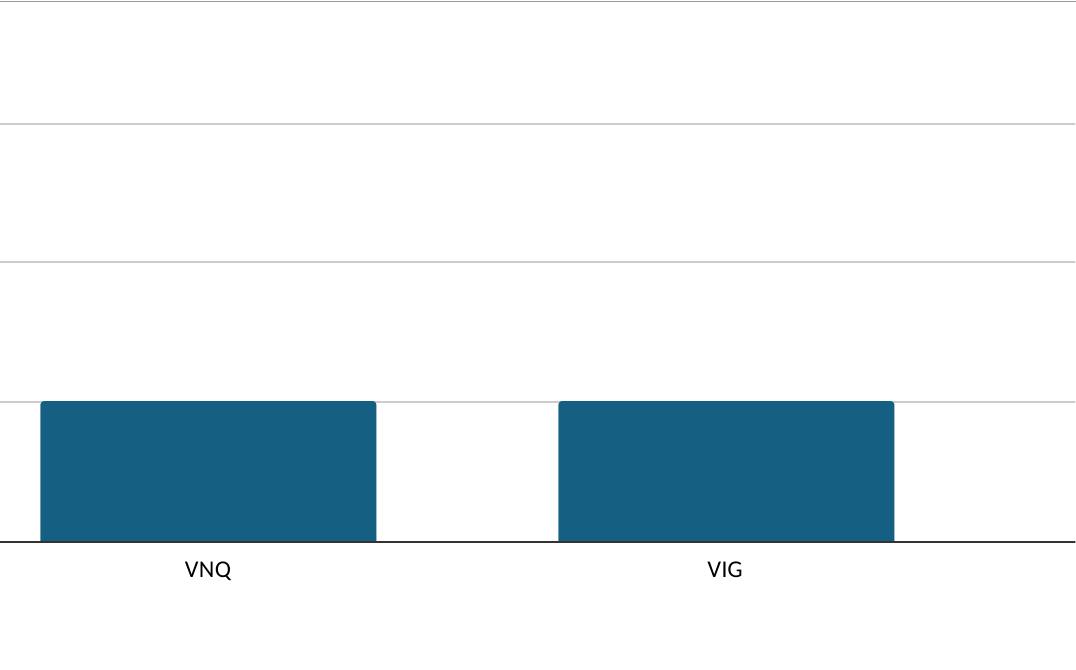
Based on the current portfolio's Sharpe ratio, volatility, drawdown, Monte Carlo downside probability, and optimizer results, I would recommend _____ because _____



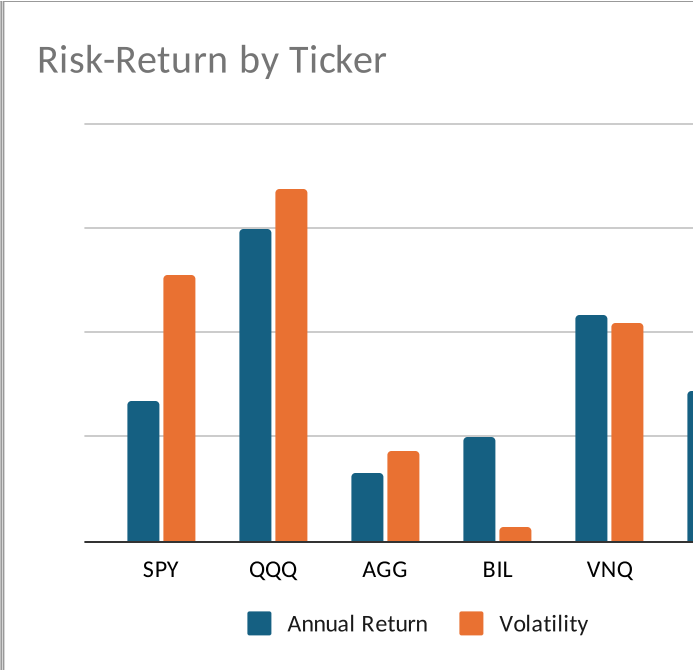
Portfolio Monthly Returns



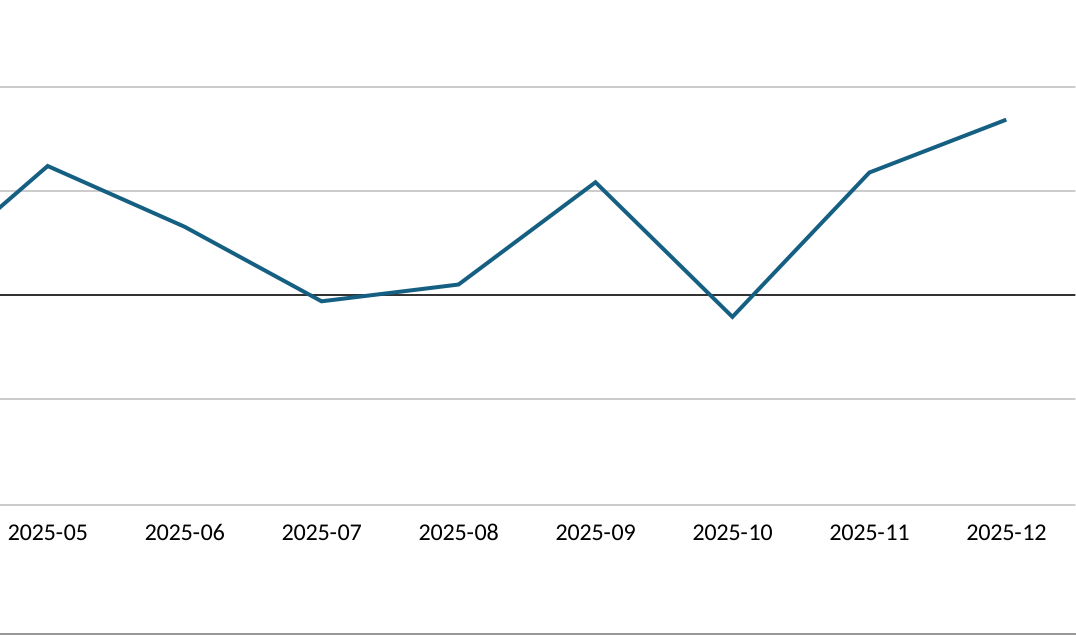
			Ticker	Allocation %		Ticker	Annual Return
\$200,000	Chart to Build	Recommended Type	SPY	30.0%		SPY	6.7%
1.2%	Allocation by Ticker	Bar/Pie	QQQ	15.0%		QQQ	15.0%
7.1%	Portfolio vs SPY Monthly Return	Line	AGG	20.0%		AGG	3.3%
-0.39x	Sharpe by Ticker	Bar	BIL	15.0%		BIL	5.0%
\$225,672	Monte Carlo Ending Values	Histogram	VNQ	10.0%		VNQ	10.8%
0.0%	Efficient Frontier	Scatter	VIG	10.0%		VIG	7.2%
	Correlation Matrix	Heatmap-style table					



Month	Portfolio Return	Month	SPY Return
2024-01	2.4%	2024-01	6.9%
2024-02	(0.6%)	2024-02	0.3%
2024-03	1.3%	2024-03	1.7%



2025-08	0.3%	2025-08	(0.6%)
2025-09	2.7%	2025-09	6.6%



2025-10	(0.5%)	2025-10	(3.1%)
2025-11	2.9%	2025-11	3.2%
2025-12	4.2%	2025-12	4.9%

Volatility

12.7%

16.8%

4.3%

0.6%

10.5%

11.1%



